

HENICHES'K, Ukraine

WMO: 339100

Lat: 46.1642N

Lon: 34.8072E

Elev: 49

StdP: 14.67

Time Zone: 2.00 (EUE)

Period: 94-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 8.5	(c) 13.4	(d) 3.4	(e) 6.6	(f) 9.4	(g) 8.3	(h) 8.4	(i) 14.4	(j) 55.3	(k) 21.6	(l) 44.5	(m) 27.2	(n) 10.9	(o) 0	(p) 0.508

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.4	88.4	72.4	86.0	71.9	84.0	70.9	75.9	84.8	74.4	83.5	72.6	81.8	7.1	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.2	123.4	82.2	71.1	114.8	80.9	69.2	107.6	79.2	39.5	84.3	37.9	83.9	36.4	81.6	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
41.0	35.9	30.3	2.7	94.3	5.4	4.2	-1.2	97.3	-4.4	99.8	-7.4	102.1	-11.4	105.1
			2.1	80.2	5.4	2.3	-1.8	81.8	-5.0	83.2	-8.0	84.5	-12.0	86.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.1	30.0	32.2	39.4	50.4	62.2	69.9	75.5	73.4	63.9	52.1	42.3	32.2
	DBStd	17.46	7.93	8.17	6.35	6.57	6.00	4.92	4.28	5.00	6.35	7.83	7.64	8.87
	HDD50	2395	620	499	333	75	2	0	0	0	1	66	247	552
	HDD65	5570	1085	918	794	439	124	15	0	3	92	404	680	1016
	CDD50	3151	0	0	4	86	381	597	790	726	419	132	16	0
	CDD65	855	0	0	0	1	38	161	325	264	60	6	0	0
	CDH74	6412	0	0	0	6	115	994	2920	2180	191	6	0	0
	CDH80	1691	0	0	0	0	9	168	853	626	35	0	0	0
(14) Wind	WSAvg	9.6	9.6	9.8	11.3	11.2	8.8	8.4	7.8	7.7	9.5	9.8	10.5	10.4
(15) Precipitation	PrecAvg	15.4	1.3	1.2	1.3	1.3	1.3	1.7	1.5	1.3	1.2	1.0	1.3	1.5
	PrecMax	24.2	3.4	3.1	3.3	3.6	4.8	8.1	5.3	4.6	6.5	3.7	2.9	3.3
	PrecMin	8.5	0.2	0.0	0.2	0.0	0.0	0.2	0.0	0.0	0.0	0.1	0.0	0.0
	PrecStd	3.7	0.9	0.7	0.7	0.8	1.0	1.4	1.5	1.2	1.1	0.8	0.7	0.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.0	52.9	61.1	72.6	81.0	87.6	93.4	92.2	84.0	73.9	62.3	52.3
		MCWB	44.7	48.6	53.1	60.6	65.4	70.0	74.7	74.1	69.9	64.2	56.6	49.9
	2%	DB	44.1	47.9	53.6	66.7	77.0	84.2	88.8	88.3	78.1	70.2	58.1	48.5
		MCWB	42.2	44.9	47.0	57.4	64.7	69.3	73.0	72.2	66.8	63.5	54.2	46.4
	5%	DB	41.5	44.0	50.4	62.7	74.4	81.6	86.4	85.6	75.6	67.2	54.8	46.0
		MCWB	40.2	41.9	45.7	55.1	64.2	68.7	72.9	71.1	66.2	62.3	51.6	44.2
	10%	DB	39.3	41.5	47.7	59.6	72.2	79.5	84.6	83.5	73.5	63.7	52.0	43.0
		MCWB	38.4	39.8	44.3	53.3	63.1	67.8	72.1	70.2	65.3	58.6	49.9	41.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.0	49.2	52.7	61.3	68.1	74.7	78.7	77.7	71.9	67.1	57.4	50.4
		MCDB	46.9	52.5	60.3	71.1	75.3	82.0	84.8	87.4	78.6	71.4	60.5	51.9
	2%	WB	42.7	45.1	48.9	58.2	67.1	72.7	76.6	75.2	69.9	65.0	54.6	46.7
		MCDB	43.9	47.5	52.2	64.8	74.1	81.1	84.9	84.1	75.0	68.9	57.7	48.3
	5%	WB	40.4	42.3	46.4	56.1	65.7	70.8	75.3	73.4	68.2	62.6	52.1	44.3
		MCDB	41.4	43.9	49.2	61.5	72.6	79.0	84.7	82.2	73.6	66.8	54.1	45.8
	10%	WB	38.3	40.1	44.6	54.3	64.1	69.3	73.7	71.7	66.8	59.2	50.1	41.8
		MCDB	39.2	41.3	47.3	58.8	71.0	77.1	83.4	81.1	72.0	63.1	51.9	42.9
(35) Mean Daily Temperature Range	MDBR		7.4	9.0	9.1	10.7	13.3	14.5	15.4	15.6	13.2	11.6	8.2	8.0
	5% DB	MCDBR	9.2	11.2	14.0	16.2	15.8	18.0	17.4	18.1	15.6	14.5	11.6	9.4
		MCWBR	7.9	8.8	9.3	9.3	8.4	9.8	9.4	9.9	8.9	9.6	8.1	8.1
	5% WB	MCDBR	8.7	10.9	12.1	13.9	13.2	15.2	14.8	15.7	12.7	12.7	10.1	8.9
		MCWBR	7.7	9.0	8.8	8.5	7.9	9.7	9.4	10.6	8.2	9.2	7.7	7.9
(40) Clear-Sky Solar Irradiance	taub		0.347	0.376	0.403	0.446	0.416	0.421	0.443	0.444	0.416	0.383	0.360	0.346
	taud		2.294	2.228	2.201	2.114	2.297	2.318	2.266	2.248	2.290	2.347	2.343	2.307
	Ebn at Noon		231	246	258	257	271	269	261	255	252	242	224	216
	Edh at Noon		26	33	39	47	40	40	41	41	36	29	24	23
(44) All-Sky Solar Radiation	RadAvg		339	606	991	1527	1951	2031	2054	1839	1328	810	412	285
	RadStd		45	83	105	137	151	145	113	100	101	91	52	45

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	+1.28	N/A	N/A	+1.76	N/A	N/A	-202	-284	+234	+149

Nomenclature: See separate page